

Beyond accuracy scores:

Distinct working memory profiles across socioeconomic and math performance levels

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1. Background

Evidence links socioeconomic status (SES) to cognitive development and academic performance (Noble & Giebler, 2020). Working memory (WM) is proposed as a mediator (Waters et al., 2020). Research often relies on global performance, which may limit the capture of cognitive diversity in developmental pathways. Exploring specific task parameters and error types could reveal distinct subgroups for targeted interventions.

2. Purpose

To implement unsupervised clustering analysis to examine associations between SES and academic performance, and to characterize WM response patterns across children with distinct profiles.

3. Methods

Fifty-one 5-year-olds completed a Corsi task and the Woodcock-Muñoz math subscale; caregivers provided SES information. Associations were examined via multiple regression and bootstrapped mediation analyses. Children were clustered (PAM), and response patterns were compared between groups using Kruskal-Wallis and corrected post-hoc tests. Analyses were conducted in R (2025).

4. Results

4.1. Regression and mediation analyses

Model A included Working memory accuracy as outcome variable, with Age and SES as predictor variables. **Model B** included Math performance and WM accuracy as well as Age as predictors. SES, WM, and math performance were associated ($\beta > 0.54$; $p < .001$), after controlling for the effect of age. Non-parametric bootstrap mediation analysis showed that working memory mediated SES's effect on math performance ($\beta = 0.21$; $p < .001$), making the direct effect disappeared.

4.2. Task parameter and Error analysis

We were interested in studying which specific aspects of performance in CORSI might be relevant in the found associations. We studied the role of the maximum achieved level, performance in trials with short, intermediate and long path lengths, and the different types of errors committed by participants (namely order errors, block errors, and other errors).

The different parameters of the task showed distinct associations with the global score in the sense that none of them alone seems to explain all changes in global score, with short and long paths showing the higher estimates. The different error types also showed such associations, with order errors having the higher estimate value.

4.3. Cluster analysis

A $k = 3$ was selected and the clustering algorithm was implemented using only SES and math score as features, resulting in the following clusters:

- Cluster 1 contains the children with Lower SES and Math score values in average
- Cluster 2 contains the children with Intermediate SES and Math score values in average
- Cluster 3 contains the children with Higher SES and Math score values in average

These groups showed statistically significant differences in maximum level, performance in short and long trials, and the incidence of order errors ($H > 7.484$; $p < .02$).

5. Conclusions

Specific response patterns were related to children's SES and math performance, which may reflect distinct developmental pathways. Results may contribute to the design of more inclusive interventions and policies that offer different activities for children with specific processing needs.

6. References

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Figure 1. Research assistants administering computerized cognitive tasks in the school.

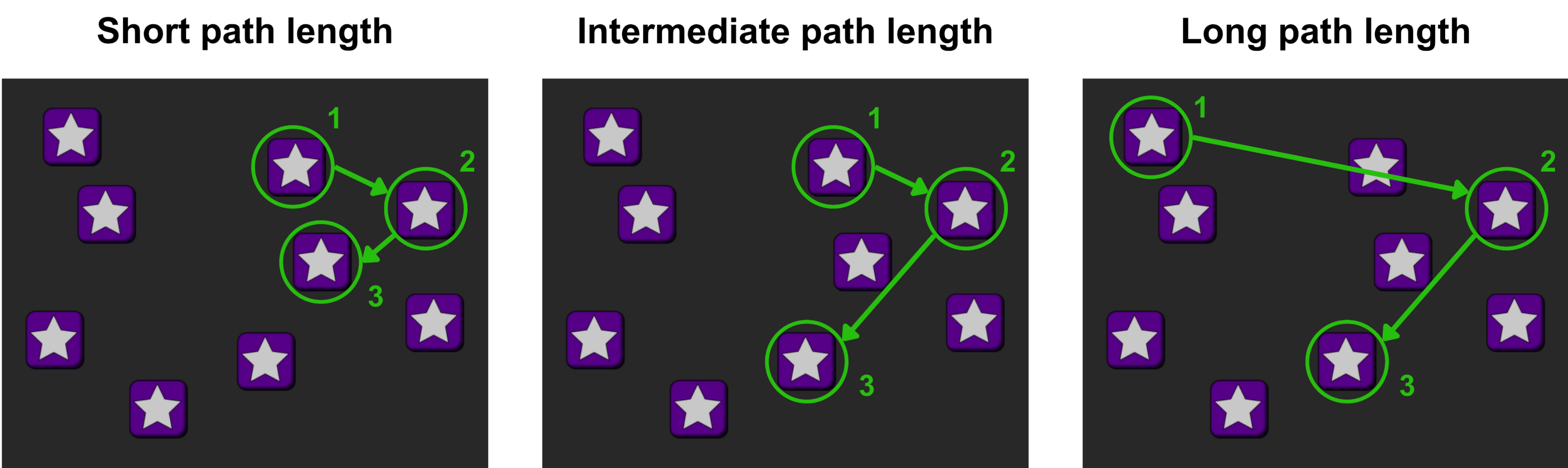


Figure 2. The three different path lengths included in Corsi task.

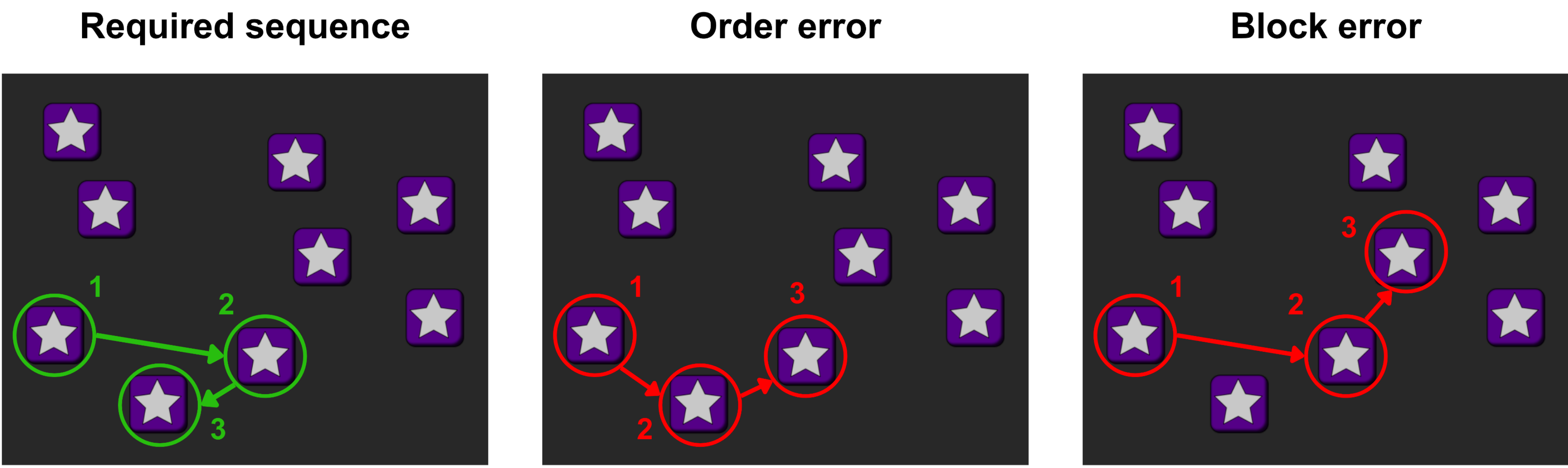


Figure 3. The three different error types differentiated from participant's responses.

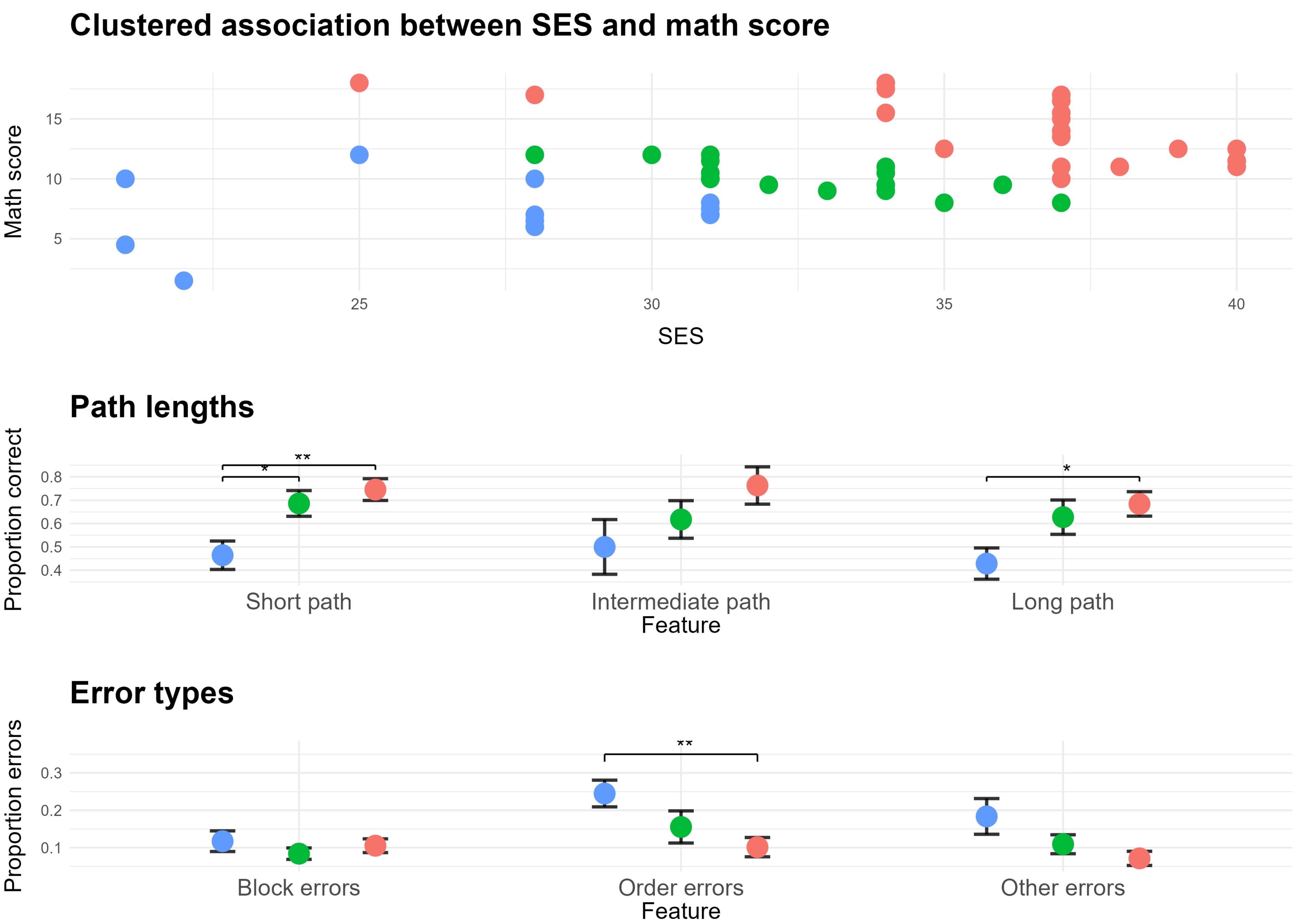


Figure 4. SES, math score, problem parameters and error types across the three distinct clusters. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

